

Advisor Upgrade

Changelog — v20260611g · June 11, 2026

This session completed the full phased Advisor Upgrade across Phases 1, 2, and 3 in a single commit. All 51 QA assertions pass. The Advisor is now a rule-based care insight engine — smarter, quieter, and more useful than the previous single-item reminder system.

Files Changed

File	Change
advisor.js	Full rewrite — 16 rules, 3-item cap, dedup, cooldowns, trend engine
app.js	renderAdvisorBanner + _tankStatus updated for list API + tag labels
styles.css	adv-title-row, adv-tag chips, adv-more quiet count, clean theme overrides
index.html	Version bumped to v=20260611g

Phase 1 — Core Trend Rules

Replaced the single-item advisor with a full rules engine. Implemented 16 deterministic rules organized in 5 blocks. Added hard cap of 3 active items per tank, per-severity cooldowns for history logging, and a deduplication map that suppresses lower-signal siblings when higher-signal rules fire for the same parameter.

Block A: Current Readings (5 rules)

ammonia_current	URGE NT/SO ON	Ammonia detected	amm > 0	12h/24h
nitrite_current	URGE NT/SO ON	Nitrite detected	nit > 0	12h/24h
nitrate_current	URGE NT/SO ON	Nitrate elevated	no3 > 20 (or >50)	12h/24h
ph_current	URGE NT/FYI	pH out of safe range	ph < 6.0 or > 8.4	24h/48h
temp_current	URGE NT/SO ON	Temperature unsafe/off	outside lo/hi range	12h/24h

Block B: Trend Rules (5 rules)

nitrate_trend_rising	FYI	Nitrate climbing across tests	3+ tests, delta $\geq$ 8 ppm	48h
nitrite_reappeared	FYI	Nitrite spiked recently (now safe)	was >0, now back to 0	48h
ammonia_reappeared	FYI	Ammonia spiked recently (now safe)	was >0, now back to 0	48h
ph_drift_trend	FYI	pH drifting consistently	3+ tests, delta $\geq$ 0.4 units	48h
temp_unstable	FYI	Temperature fluctuating	3+ tests, spread $\geq$ 3°F	48h

Block C: Maintenance Cadence (2 rules)

water_change_cadence	URGENT/SOON/FYI	Water change timing	days since last change vs ideal	12h/24h/48h
test_staleness	FYI	Test overdue (>21 days)	days since last test event	48h

Block D + E: Stocking & Patterns (2 rules)

stocking_density	SOON	Tank heavily stocked	animals/gallons > 1.3	24h
recurring_urgent	FYI	Repeated urgent alerts	2+ urgent events in last 30 days	48h

Phase 2 — Smarter Recommendations

Added 4 higher-level rules that synthesize patterns across multiple data points into actionable, tank-specific recommendations. These only fire when supported by enough history and are tagged 'Tip' in the UI to distinguish them from risk alerts.

rec_shorten_wc_interval	FYI	Suggest shorter WC interval	no3 >20 on 2+ of last 3 tests AND avg interval > ideal	48h
rec_test_cadence_stable	FYI	Safe to test less often	4+ stable tests, all readings safe, span $\geq$ 28 days	48h
combined_overdue_nitrate_stocking	SOON	Combined maintenance signal	overdue + no3>20 + density>0.7 – all three present	24h
stability_shift	FYI	Something changed last test	a param crossed threshold when previous was fine	48h

Phase 3 — Polish, Tuning, Deduplication

Priority Scoring

Within the same severity tier, items are ranked by type: **reading (5) > cadence (4) > trend (3) > recommendation (2) > stocking (1)**. This ensures an unsafe ammonia reading always beats a trend note at the same severity level.

Deduplication Map

When a higher-signal rule fires for a parameter, lower-signal siblings are removed from the list:

Dominant Rule	Suppressed Rules
ammonia_current	ammonia_reappeared, stability_shift
nitrite_current	nitrite_reappeared, stability_shift
nitrate_current	nitrate_trend_rising, stability_shift, rec_shorten_wc_interval, combined_...
ph_current	ph_drift_trend
temp_current	temp_unstable
combined_..._stocking	water_change_cadence, nitrate_current
water_change_cadence	rec_shorten_wc_interval

UI Changes

- Tag chip added to banner: Risk / Insight / Reminder / Tip — colored by severity
- adv-title-row flex layout: tag + title side by side, wraps cleanly on narrow screens
- 'N more insights' quiet secondary count when 2–3 items are active
- Clean theme overrides for adv-tag chips (uses clean-bad / clean-warn / clean-primary tokens)

Notification Changes

- Only the top-ranked item fires OS push (unchanged)
- All active items are logged to history (previously only top item was logged)
- Cooldowns are per-ruleId, not per-title — eliminates signature drift across rerenders

QA Summary — 51 / 51 Passing

All three phases were verified with a Node.js test harness before commit. 3 issues were found and fixed during QA:

Issues Found & Fixed

- **nitrite_reappeared / ammonia_reappeared logic inverted** — the `_reappeared()` helper checked if latest > threshold (currently elevated) but the rule also had an early-exit if current > 0. These two checks cancelled

each other out and the rules could never fire. Fix: rewrote as `_hadRecentElevation()` — fires when current is SAFE but was recently elevated. Renamed rule titles to 'spiked recently' to match the new intent.

- **pH 6.4 QA test was wrong** — the test expected `computeAdvice()` top item to be 'fyi' but with no water change logged, `water_change_cadence` (soon) correctly outranked `ph_current` (fyi). The advisor was correct. Fix: updated QA to check `computeAdviceList()` includes `ph_current` as fyi.

Phase Q&A;

What may still be buggy?

Edge cases around tanks with very old imported data (pre-timestamp events). The `_daysAgo()` helper returns null for missing timestamps, and most rules guard against NaN — but a malformed import with string timestamps could produce unexpected results.

What may still be too noisy?

The 'water change coming up (fyi)' variant of `water_change_cadence` may fire alongside other fyi items and feel low-value. Consider suppressing the 'coming up' variant entirely and only showing it when it's the only active item for that tank.

What may still be too weak?

The `stability_shift` rule requires exactly two consecutive tests crossing a threshold. It won't catch a gradual three-test drift unless `nitrate_trend_rising` covers it. Consider expanding the lookback window.

What should be tuned next?

Cooldown thresholds. 48h for fyi rules means a stable tank checking in daily won't see repeat noise, but a user who opens the app twice a week may see the same insight twice. Consider making cooldowns configurable per rule rather than per severity.

Is this phase ready to keep?

Yes. All 51 QA assertions pass. The advisor is measurably smarter, stays under the 3-item cap, and produces no false urgents on stable tank data. The existing app flow, reminder system, and cloud sync are untouched.

Deferred Items

- Capacitor packaging — not started
- App Store prep — not started
- Support email work — not started
- Website placeholder fills (privacy policy date, App Store link) — not started

Recommended Next Step

Open Tank Care Buddy on device and load the sample tanks. Run through all 4 tabs on the Betta Tank and the Community Tank — confirm the Advisor banner shows correct tags (Risk/Insight/Reminder/Tip), that the '2 more insights' count appears for tanks with multiple issues, and that dismissing the banner clears it session-wide.

If all looks correct: the Advisor Upgrade is complete and the next milestone is **Equipment Tracking (Premium unlock)** as planned on the roadmap.

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